

Pre-Lab Plan Guidelines and Template

Each laboratory experiment has a well-defined objective. This objective is accomplished through data taking, reduction, and analysis. These tasks may be successfully completed and the objective achieved only when the nature of the desired result is known, before the experiment is undertaken. Otherwise insufficient data may be taken, the results may be incorrectly analyzed, or preposterous results may be reported.

Read the lab manual for your current experiment. Familiarize yourself with the theory relevant to the experiment and the equipment necessary to take the required data. Read the appropriate sections of the Unit Operations textbook and supplemental information in Blackboard and other resources such as Perry's Handbook. When you are sure you understand the objective of the lab, prepare in writing, a Pre-Lab Plan. This exercise is intended to provide the preparation necessary to undertake a successful experimental study. Most important will be understanding the objective of the experiment, familiarization with the apparatus and materials/supplies needed to carry out the experiment, and a detailed safety analysis focusing on the hazards involved.

The template for the Pre-Lab Plan is given below. The plan consists of a Title page (1 page), Experimental section (2-3 pages) and a Safety Analysis sheet (1 page) for a total of 4-5 pages. Follow this template when writing your Pre-Lab Plan.

You will be required to electronically submit your Pre-Lab Plan at the end of the Pre-lab period. Submit as a Word document and use the following filename format:

PreLab#TimeDayTeam#.docx

For example: PreLab1AMTuesdayTeam3.docx

(Lab Title)

Pre-Lab Plan
Unit Operations Lab #
Date

Group #, Tuesday/Thursday, AM/PM section

Team member Names

Name 1

Name 2

...

1. Experimental Objective (5 pts)

Briefly introduce the objective of the lab and its technical nature (1-2 sentences). This information comes from the problem description, or other context of the assignment.

2. Experimental

This section must be clear, concise and specific, and its specific contents will depend on the actual experimental plan you have designed. Topics covered within this section must include the following:

2.1 Apparatus (5 pts)

Describe the system under investigation; the main and auxiliary apparatus used. Be specific but brief and concise and include a process flow diagram or picture of the apparatus. For clarity, a table should be used to list the important components with a brief description of each. Number the components and show where they are located on your diagram.

2.2 Materials and Supplies (5 pts)

Use a table to describe your materials and supplies. Remember that materials and supplies are different than your equipment.

2.3 Experimental Plan (10 pts)

Outline the experimental plan that you have designed. Do not just copy/paste from the lab manual, but write it in your own words. Include the experimental procedure, and the number, order, conditions and other details of the experimental runs that will be conducted. State what the independent and dependent variables are, and how and over what range the variables will be measured. Statistical and/or calibration methods used to estimate/reduce experimental uncertainty can be described here. Be specific but brief. How many replicates? What confidence interval? How will goodness of fit be measured? The plan should also describe how the data will be correlated with theory, empirical equations or charts, or engineering models.

Project Safety Evaluation (10 pts)

Project Title: _____

Date: _____

	<i>Yes/No</i>	
Potential electrical hazards?		If yes, identify conditions:
Potential mechanical hazards?		If yes, identify conditions:
Potential pressure hazards?		If yes, identify conditions:
Potential temperature hazards?		If yes, identify conditions:
Hazardous/toxic chemicals involved?		If yes, hazards involved:
Gloves required?		If yes, specify type:
MSDS reviewed by all team members?		List MSDS sheets reviewed:
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)		
Pressure (psia)		
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>

I have read relevant background material for the Unit Operations Laboratory entitled: “_____” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

signature of team member date

signature of team member date

signature of team member date

signature of team member date